

BUSS386 FUTURES AND OPTIONS

Fall 2026

Course Information

- Time: Tuesdays, 3:00–5:45 pm
- Classroom: LP 433
- Instructor: Ji-Woong Chung (chung.jiwoong@korea.ac.kr)
- Teaching Assistant: Seunghwan Lee (seunghwan514@korea.ac.kr)
- Office Hours: By appointment via email

Course Website: <https://chung-jiwoong.github.io/BUSS386>

Course Description This course covers the fundamentals of derivative securities—forwards, futures, options, and swaps—and their applications in pricing, hedging, and risk management.

Learning Objectives By the end of the course, you should be able to:

1. Describe the mechanics and market structure of forwards, futures, options, and swaps, and explain who uses each instrument and why.
2. Price forwards and futures by no-arbitrage (cost-of-carry) and value interest-rate and currency swaps by replication.
3. Build and interpret option payoff and profit diagrams, and design trading strategies (spreads, straddles, collars) to express a market view.
4. Apply put-call parity and no-arbitrage bounds to detect and exploit mispricing.
5. Price options with the binomial and Black–Scholes–Merton models, and measure and hedge risk using the Greeks.
6. Interpret implied volatility, the volatility smile and surface, and the VIX using real market data.

Prerequisite Recommended: BUSS207 (Financial Management). Comfort with basic algebra, probability, and the time value of money is assumed.

Main References (supplementary)

- Robert L. McDonald, *Derivatives Markets*.
- John C. Hull, *Options, Futures, and Other Derivatives*.

Grading

Component	Weight
Midterm Exam	40%
Final Exam	50%
Class Participation	10%
Total	100%

Notes on Grading

- Both exams are closed-book. One A4 hand-written cheatsheet (both sides) is allowed; typed sheets are not permitted.
- No glasses, sunglasses, or eyewear of any kind may be worn during examinations: Students who require prescription eyeglasses to read the exam must present their glasses to the proctor for physical inspection prior to the start of the exam. Uninspected eyewear will not be permitted under any circumstances.
- Midterm: Tuesday, October 20, 15:10–17:10. Final: Tuesday, December 15, 15:10–17:10.
- The final is cumulative, with emphasis on material after the midterm.
- Class participation reflects engagement in class and on the online board; attendance will be randomly checked.
- Makeup exams are granted only for documented emergencies (illness, family, or university-sanctioned absence), with advance notice whenever possible.
- Error Bounty: If you find an error (typo, computational mistake, or unclear phrasing) in the problem sets or solutions, notify the instructor or TA with a brief description. Each valid error reported earns 0.5 bonus points toward your overall grade.

Course Outline (tentative; dates adjust to the Korean academic calendar and any announced holidays)

Week	Date (Tue)	Topic
1	Sep 1	Introduction to derivatives
2	Sep 8	Interest rates, yield curves
3	Sep 15	Forwards & futures pricing
4	Sep 22	<i>No class — Chuseok holiday</i>
5	Sep 29	Hedging with futures
6	Oct 6	Swaps
7	Oct 13	Options: intro, payoffs, properties & put-call parity
8	Oct 20	Midterm Exam
9	Oct 27	Option trading strategies
10	Nov 3	Binomial trees
11	Nov 10	Black–Scholes–Merton
12	Nov 17	Greeks and delta hedging
13	Nov 24	Volatility: smile, surface, VIX
14	Dec 1	Exotic options
15	Dec 8	Review
16	Dec 15	Final Exam

Other Policies

- **Generative AI:** AI tools may be used as a study aid—clarifying concepts, checking derivations, or debugging spreadsheet formulas—and any such use should be acknowledged. Exams are closed-book and individual; submitting AI-generated work as your own on graded assessments is academic misconduct.
- **Academic Misconduct:** Cheating and plagiarism will result in course failure and formal reporting to the university committee.
- **Grading Appeals:** Appeals must be submitted in writing (email acceptable) within 5 days of grade release and must state clear grounds.
- **Students with Disabilities:** Inform the instructor immediately to arrange accommodations.
- **Copyright:** Course materials are for personal study only. Recording or redistribution without permission is prohibited.
- **Anti-Graft Law:** Requests for grade changes without valid grounds are prohibited under the Improper Solicitation and Graft Act.